Network Science Data II
PHYS 7332, Fall 2025

Prof. Brennan Klein

Assignment 3, due Friday, Nov. 21, 11:00pm ET.

Please submit a .pdf of your answers via Canvas, including a url to your .ipynb notebook in your fork of our class repository. Please name your Canvas submission as Lastname.pdf, (*e.g.* Klein.pdf). If you need help with LaTeX or GitHub, please visit us during office hours or post questions to our email thread.

1. *Community detection, degree heterogeneity, and* `graph-tool` *(45 points).*

   This question is a deeper dive into the inferential approach to network science, focusing specifically on the relationship between degree heterogeneity and block structure.

   (a) Recall in `class_09`, we looked at modularity in random (i.e., $G(n, m)$) graphs. Expand this result by replicating Figure 2 from Guimera, R., Sales-Pardo, M., & Amaral, L.A.N. (2004). Modularity from fluctuations in random graphs and complex networks. *Physical Review E*, 70(2), 025101. Comment on this result, especially as it relates to modularity as a measure of partition quality.

   (b) Now, using `graph-tool`, fit a degree-corrected and non-degree-corrected stochastic block model to the largest connected component of the `polblogs` dataset, and enforce that the resulting partitions only contain two blocks. Visualize the graph (twice) with the two different partitions. Hints: 1) Access this dataset using Netzschleuder, by calling `gt.collections.ns["polblogs"]`, 2) You can run then `g = gt.extract_largest_component(g, directed=False)`, and 3) The function `minimize_blockmodel_dl` takes multiple keyword arguments, which you will need to specify to complete this question.

   (c) Let's dive more into the Markov Chain Monte Carlo (MCMC) algorithm used to arrive at the partitions when fitting stochastic block models.

      i. Print the description length of the degree-corrected and non-degree-corrected models you fit in (b).

      ii. Fit new degree-corrected and non-degree-corrected models to the largest component of the `polblogs` dataset. After initializing `BlockState` with `B=2`, use a `for` loop of 1,000 iterations to collect the description lengths of each model at every step using the `mcmc_sweep` function (setting `niter=1`; during the for-loop *do not enforce the number of blocks to be 2*). Visualize both of these curves in a single plot, labeled properly, and comment on your observation.

   (d) Explore the **model selection** section in the `graph-tool` documentation and Section VIIA and VIIB in `https://arxiv.org/abs/1705.10225`. Figure 7 in this article should validate whether your approach in (b) above is on the right track. Summarize the intuition and formalism behind degree correction in the (microcanonical) stochastic block model framework. Use Figure 8 as your guide.

2. *Core concepts in machine learning and/or network science. (25 points).*

Learning machine learning and network science can be tough, and sometimes visualizations can help build intuition (either the process of making them, or the process of walking through one). This question asks you to build a *pedagogically effective* visualization of a concept in machine learning or network science. [Note: Reach out to Brennan if you are unsure about topic selection.]

(a) Pick a machine learning algorithm or network science technique that was not covered in class. In your words, summarize how it works and what it's useful for.

(b) What aspect of the model is most confusing to you? Explain why it is confusing conceptually and walk us through your own understanding in whatever format works best for you.

(c) Create a flashcard for the difficult concept you just walked us through. Use `matplotlib`, LaTeX, and any other illustrative techniques that are appropriate. [*Potential 5 bonus points if this is made entirely in Python (e.g. Matplotlib, etc.)*]

3. *Random walk dynamics. (15 points).*

In `class_16`, we introduced the concept of the mean *first-passage time* of random walkers on a graph. Using Masuda, N., Porter, M. A., & Lambiotte, R. (2017). Random walks and diffusion on networks. *Physics Reports*, 716, 1-58. as your guide:

(a) Write a function from scratch that computes the *commute time* of a single random walker between nodes i and j in a network. Test this on the `karate_club_graph` by printing the commute time between three randomly selected pairs of nodes.

(b) Calculate the *mean* commute time between pairs of nodes in Erdős-Rényi networks with $N = 500$ and $\langle k \rangle = 8$ and Barabási-Albert networks with $N = 500$ and $m = 4$. Compute the average commute time across all nodes in the network, then average this value across 10 samples from each of these two graph models. Comment on any differences you observe.

4. *Meta-question about random walks and spectral properties. (15 points).*

This question flips the tables. Create a problem set question that has students learn about the connection between random walk network dynamics and *a spectral property of your choice*. This could focus on the graph Laplacian, the effective resistance, the non-backtracking matrix, or any spectral property of interest. The question should be multi-part, and it should build on itself. The question should incorporate some coding, but it does not need to be entirely a computational exercise. If you have questions about your selection and how your question is structured, please contact me during office hours.